SYSTEM REQUIREMENT DOCUMENT (SRD)

Requirement	[eAuto] New UCD: Enforce 12-Character Password Complexity Across All Portals with uppercase, lowercase, number & special character
Version	Version 1.0
Latest Revision Date	26.08.2026 (Wednesday)

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1.0 DOCUMENT INFORMATION

1.1 Purpose

The System Requirements Document outlines the agreed system specifications. Once approved, it becomes the baseline to ensure the system meets business goals. Sign-off is required for development to start, and any changes must follow the System Enhancement Management process.

1.2 Intended Audience

The intended audience for this document includes project stakeholders and any team members involved in defining, implementing, or validating the system requirements.

1.3 Revision History

This section records all modifications to the document until it reaches Version 1.0, at which point it is finalized and established as the baseline for future reference. Once baselined, any new changes will be assessed based on their severity — significant changes may result in an addendum, while less critical updates may be handled through a formal change request.

Version No	Date	Revised By	Reason for Revision
0.1	24.08.2026 (Mon)	Lim Yi Link	Initial Draft
1.0	26.08.2026 (Wed)	Lim Yi Link	Send out for external reference

1.4 Glossary of Terms

#	Abbreviation / Term	Interpretation / Description
1	UCD	Used Car Dealer
2	BO	BackOffice
3	AATF	Authorised Automotive Treatment Facility
4	Main User	The primary account holder of a UCD company, able to manage sub-user accounts
5	Sub User	A secondary user account created by the Main User under the same company
6	MD5	The hashing algorithm applied by the browser before the password is submitted on selected pages
7	ASCII	The standard printable English character set
8	Special Character	A character from the approved set: ! @ # $ % ^ & * ( ) _ + - =

1.5 Associated Document(s)

#	Item	Location / URL	Rationale for Relevance
1	EAINT-12155 — Enforce 12-Character Password Complexity Across All Portals with uppercase, lowercase, number & special character	https://mfservices.atlassian.net/browse/EAINT-12155	Source enhancement request for this SRD
2	CCB-899 — EAUTO | To Enforce 12-Character Password Complexity Across All Portals (uppercase, lowercase, number & special character)	https://mfservices.atlassian.net/browse/CCB-899	Change Control Board ticket for this enhancement
3	EAINT-12220 — Password-reset endpoints have no server-side role check	https://mfservices.atlassian.net/browse/EAINT-12220	Related security defect on the same password-reset endpoints; handled separately

2.0 REQUIREMENT

2.1 High-Level Requirement Overview

#	Item	Description
2.1.1	Enhancement Overview	Strengthen the eAuto password policy and apply it to newly created accounts across all portals. Every password set for a new account, whether entered by an administrator or system-generated, must be at least 12 characters and must contain an uppercase letter, a lowercase letter, a number, and a special character.
2.1.2	Key Value(s) Expected	1. Reduce the risk of account compromise caused by weak or short passwords. 2. Remove the inconsistent password rules currently duplicated across multiple classes. 3. Close the flows that today accept a password without any server-side validation. 4. Replace weak system-generated passwords with policy-compliant ones.

#	Item	Description
2.1.3	Additional Note(s)	The policy applies to accounts created on or after the deployment date only. Existing accounts are exempt. Stored passwords remain valid, no forced password migration is performed, and an existing user is never required to change a password that does not meet the new policy.

2.1.1 Assumptions

#	Assumption
1	Existing stored passwords remain valid after deployment. No forced password migration is carried out for current users, and an existing user is never prompted to change a non-compliant password.
2	A single validation utility is introduced as the source of truth for the password policy, and all server-side call sites delegate to it.
3	The My Account password change and the Change Main User Password pages hash the password in the browser before submission, so the server never receives the plaintext. Client-side validation is the accepted enforcement point for these two pages.
4	All other password flows submit the plaintext password to the server and can therefore be validated server-side.
5	The client-side and server-side hashing implementations agree only on printable ASCII input, so non-ASCII characters must be blocked to avoid permanent lockout.
6	The system can distinguish a newly created account from an existing account, so that the policy is applied to new accounts only. The mechanism for this distinction is to be confirmed with the development team.

2.1.2 Out-of-Scope

#	Item	Reason
1	Password expiry and rotation policy	Not requested under this enhancement
2	Password history check to block reuse of previous passwords	Not requested under this enhancement
3	Server-side role check on the password-reset endpoints	Handled separately under EAINT-12220
4	Migration of existing non-compliant passwords	Accounts existing before the deployment date are exempt by agreed business decision
5	Two-factor authentication	Not part of this requirement
6	Password change pages used by existing users, covering My Account, Update User Account, Change Main User Password and Change Director Password	The policy applies to newly created accounts only. Existing accounts retain the current password rules.
7	Password reset of an existing BackOffice user by an administrator	The reset issues a password to an existing account, which is exempt from the new policy.

#	Item	Reason
8	Forgot password temporary password for an existing user	The temporary password is issued to an existing account, which is exempt from the new policy.

2.2 Key Highlights and Baseline Requirements

The following key highlights and requirements were established during the initial sync-up sessions with the requestor. As this forms the baseline, any changes requested thereafter may result in adjustments to scope, timeline, or deliverables, for which the requestor will be responsible to review and confirm.

2.2.1 Background

#	Affected Item	Description
1	Project	eAuto
2	Portal	UCD, BackOffice, JPJ, AATF, Bank and Insurance
3	Module	User Account and Password Management
4	Request	Enforce a single 12-character password policy across all portals whenever a password is set, changed, or generated.
5	Additional Note	Accounts existing before the deployment date are exempt. The policy applies to accounts created on or after that date.

2.2.2 Existing Issue & Expected Behaviour

#	Item	Description
1	Existing Issue / Behaviour	1. The minimum length is 6 characters, and 8 for AATF and Insurance. No uppercase, lowercase, number or special character is required. 2. The rule is duplicated across several classes and two copies contain a logic error that lets almost any password pass. 3. Several flows have no server-side validation, including company creation and the AATF company pages. 4. System-generated passwords are weaker than the policy itself. 5. A password containing & = + % # corrupts the login request parameter.
2	Expected New Handling / Behaviour	1. One policy applies to every portal: minimum 12 characters with at least one uppercase letter, one lowercase letter, one number and one special character. 2. Spaces are rejected and only printable English characters are accepted. 3. A single validation utility is introduced and every server-side flow delegates to it. 4. System-generated passwords comply with the policy. 5. A password containing special characters is accepted correctly at login.

2.2.3 Affected Pages

#	Affected Pages	Description
1	UCD Portal — Login	Accepts a password containing special characters without corrupting the request parameter
2	UCD Portal — First-Time Login Password Change	Enforces the new policy when a newly created user sets the first password
3	UCD Portal — Settings > User Accounts Listing > Add User Account	Enforces the new policy when the Main User creates a new sub-user account
4	BackOffice — Manage BackOffice User Account > Add User	Enforces the new policy when a new BackOffice user account is created
5	BackOffice — Company Account Creation	Enforces the new policy on the initial director password of a newly created company
6	BackOffice — AATF Company Create	Enforces the new policy when a new AATF company account is created
7	Service Portal — Login	Accepts a password containing special characters without corrupting the request parameter

#	Affected Pages	Description
8	Service Portal — First-Time Login	Enforces the new policy when a newly created Service Portal user sets the first password

2.2.4 Password Policy Rules

The rules in this section apply when a password is set for a newly created account. Accounts existing before the deployment date are exempt.

#	Requirement ID	Item	Description	Acceptance Criteria
1	EAINT-12155-REQ-001	Existing Account Exemption	The password policy shall apply to accounts created on or after the deployment date only. The existing stored passwords shall remain unchanged. Upon login, an existing user shall be able to sign in with the current password even if it does not meet the new policy, and shall not be prompted to change it.	An existing account with a legacy password logs in successfully after deployment, is not forced to change the password, and is not shown any policy message.

2.2.5 UCD Portal — Password Change Pages

#	Requirement ID	Item	Description	Acceptance Criteria
1	EAINT-12155-REQ-002	First-Time Login Password Change	The existing forced password change on first login shall remain unchanged. Upon a newly created UCD user setting the first password, the system shall enforce the new password policy.	A non-compliant password is rejected with the relevant per-rule messages. A compliant password is accepted and the user can log in again with it, including when it contains special characters.
2	EAINT-12155-REQ-003	Confirm Password Match	The existing confirm password field shall remain unchanged. Upon the Confirm Password value not matching the new password, the system shall reject the change.	A mismatched Confirm Password value is rejected and the password is not updated.
3	EAINT-12155-REQ-004	Policy Hint Text	The existing password hint text shall be replaced. Every page on which a password is set for a new account shall display the new policy.	Every password field used to set a password for a new account displays the 12-character policy. Pages used by existing users are unchanged.

2.2.6 BackOffice Portal — Password Change and Reset Pages

#	Requirement ID	Item	Description	Acceptance Criteria
1	EAINT-12155-REQ-005	Company Account Creation	The existing company creation flow shall remain unchanged. Upon the initial director password of a newly created company being submitted, the system shall enforce the password policy.	A non-compliant initial director password is rejected and the company account is not created.
2	EAINT-12155-REQ-006	Per-User Create and Update	The existing user creation flow shall remain unchanged. Upon a password for a newly created user account being submitted, the system shall enforce the password policy. The update flow for an existing user account is excluded.	A non-compliant password for a new user account is rejected and the user record is not saved. An existing user account is not affected.
3	EAINT-12155-REQ-007	Deregistration and AATF Company Pages	The existing AATF company create flow shall remain unchanged. Upon a password for a newly created AATF company being submitted, the system shall enforce the password policy. The update flow for existing deregistration account are excluded.	A non-compliant password submitted from the AATF company create page is rejected.

#	Requirement ID	Item	Description	Acceptance Criteria
4	EAINT-12155-REQ-008	Server-Side Enforcement	Upon a password being submitted directly to the server without the browser validation running, the system shall apply the same policy and reject a non-compliant password.	A direct submission with a non-compliant password is rejected on every server-validated page. The My Account and Change Main User Password pages are excluded, as the server receives only the hashed value.

2.2.7 System-Generated Passwords

#	Requirement ID	Item	Description	Acceptance Criteria
1	EAINT-12155-REQ-009	Forced Change for Generated Password	The existing forced password change on first login shall remain unchanged. Upon logging in with a generated password, the sub-user shall be required to set a new compliant password.	The sub-user cannot access any other page until a compliant password is set.

2.2.8 Bank and Insurance Portals

#	Requirement ID	Item	Description	Acceptance Criteria
1	EAINT-12155-REQ-010	Bank Portal First-Time Change	The existing Bank portal first-time password change shall remain unchanged. Upon a new password being submitted, the system shall validate it on the server against the new policy.	A non-compliant password is rejected on the Bank portal first-time change page.
2	EAINT-12155-REQ-011	Insurance Portal First-Time Change	The existing Insurance portal first-time password change shall remain unchanged. Upon a new password being submitted, the system shall validate it on the server against the new policy.	A non-compliant password is rejected on the Insurance portal first-time change page.
3	EAINT-12155-REQ-012	Insurance Minimum Length Alignment	The existing Insurance minimum length of 8 characters shall be replaced by the common 12-character policy for newly created Insurance accounts.	An Insurance password of 8 to 11 characters set for a newly created account is rejected after deployment.

2.2.9 Password Field Validation Specification

The Password and Confirm Password input fields on every affected page shall follow the specification below.

#	Spec	Value
1	Min Length	12 characters
2	Max Length	30 characters
3	Alphanumeric	Yes — at least one uppercase letter, one lowercase letter and one number
4	Space	Not allowed
5	Special Character	Yes — at least one from ! @ # $ % ^ & * ( ) _ + - =
6	Only Allow English	Yes
7	Mandatory / Optional	Mandatory when a password is set for a new account

The following table lists the validation scenarios and the message displayed to the user.

#	Scenario	Expected Behaviour	Error Message
1	Below 12 characters on save	Password is not saved	Password must be at least 12 characters long.
2	30 characters reached	No additional character allowed to enter	—
3	No uppercase letter	Password is not saved	Password must contain at least one uppercase letter (A–Z).
4	No lowercase letter	Password is not saved	Password must contain at least one lowercase letter (a–z).
5	No number	Password is not saved	Password must contain at least one number (0–9).
6	No special character	Password is not saved	Password must contain at least one special character (! @ # $ % ^ & * ( ) _ + - =).
7	Space or non-English character entered	Password is not saved	Password must not contain spaces and must use English characters only.

#	Scenario	Expected Behaviour	Error Message
8	Confirm Password does not match	Password is not saved	Confirm Password does not match the new password.
9	All rules met	Password is saved and a green tick is displayed beside the field	—

2.2.10 Error Handling

#	Requirement ID	Item	Description	Acceptance Criteria
1	EAINT-12155-REQ-013	Non-Compliant Password	Upon a password failing one or more policy rules, the system shall display a message for each failed rule and shall not save the change.	Example 1: Password shorter than 12 characters → Password must be at least 12 characters long. Example 2: Password missing an uppercase letter → Password must contain at least one uppercase letter. Example 3: Password missing a special character → Password must contain at least one special character.
2	EAINT-12155-REQ-014	Confirm Password Mismatch	Upon the Confirm Password value not matching the new password, the system shall display a mismatch message and shall not save the change.	The message Confirm Password does not match the new password is displayed and the password is not updated.

#	Requirement ID	Item	Description	Acceptance Criteria
3	EAINT-12155-REQ-015	Invalid Character Entered	Upon a space or a non-English character being entered, the system shall reject the password and display the character restriction message.	The message Password must not contain spaces and must use English characters only is displayed and the password is not updated.

APPENDIX — REQUIREMENT CHANGE LOG

#	SRD Version	Requirement ID	Status	Reason
1	0.1	EAINT-12155-REQ-001	Added	Initial requirement for SRD v0.1
2	0.1	EAINT-12155-REQ-002	Added	Initial requirement for SRD v0.1
3	0.1	EAINT-12155-REQ-003	Added	Initial requirement for SRD v0.1
4	0.1	EAINT-12155-REQ-004	Added	Initial requirement for SRD v0.1
5	0.1	EAINT-12155-REQ-005	Added	Initial requirement for SRD v0.1
6	0.1	EAINT-12155-REQ-006	Added	Initial requirement for SRD v0.1

#	SRD Version	Requirement ID	Status	Reason
7	0.1	EAINT-12155-REQ-007	Added	Initial requirement for SRD v0.1
8	0.1	EAINT-12155-REQ-008	Added	Initial requirement for SRD v0.1
9	0.1	EAINT-12155-REQ-009	Added	Initial requirement for SRD v0.1
10	0.1	EAINT-12155-REQ-010	Added	Initial requirement for SRD v0.1
11	0.1	EAINT-12155-REQ-011	Added	Initial requirement for SRD v0.1
12	0.1	EAINT-12155-REQ-012	Added	Initial requirement for SRD v0.1
13	0.1	EAINT-12155-REQ-013	Added	Initial requirement for SRD v0.1

#	SRD Version	Requirement ID	Status	Reason
14	0.1	EAINT-12155-REQ-014	Added	Initial requirement for SRD v0.1
15	0.1	EAINT-12155-REQ-015	Added	Initial requirement for SRD v0.1